

Title: Automated Ball & Cup Toss Game

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Description:

The **Automated Ball & Cup Toss Game** is a **microcontroller-based** interactive game that challenges players to toss balls into **moving cups**. The system utilizes **capacitive sensors** to detect successful shots, with real-time scoring, feedback mechanisms, and optional difficulty adjustments. The game can be enhanced with **LED indicators, sound effects, and a timer**, making it engaging for both casual and competitive play.

Features:

1. **Moving Cups:** Cups are mounted on a **motorized platform** that moves back and forth, making the game dynamic and challenging.
2. **Capacitive Sensor-Based Detection:** **Capacitive sensors** inside each cup detect when a ball lands, ensuring accurate and instant scoring.
3. **Scoring System:** Every successful shot is automatically counted, and the score can be displayed on an **LED screen, LCD screen, LED matrix, or serial monitor**.
4. **Sound & Light Feedback:** When a ball lands in a cup, a **buzzer sounds** and an **LED light** flashes to confirm the hit. Different colors or tones can be used for each cup.
5. **Time-Based Challenges:** Players can compete within a **fixed time limit** (e.g., 60 seconds) to achieve the highest score.
6. **Multiple Difficulty Levels:** The game can be customized by **adjusting cup speed, changing cup sizes, or adding obstacles** to increase difficulty.
7. **Enhanced User Experience:** The interactive gameplay, combined with **visual and audio feedback**, ensures an engaging experience for both players and spectators.
8. **Portability & Low Power Consumption:** The entire system is designed to be **compact and energy-efficient**, making it easy to set up in different locations.